TYLER W. HUGHES

Differentiable Simulation · Inverse Design · Agentic Systems for Science

Senior Technical Lead · Flexcompute Inc. · New York, NY

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ABOUT

I build physics simulation software that engineers (and increasingly, AI agents) use to design real hardware.

EXPERIENCE

Senior Technical Lead Sep 2019 – present

Flexcompute Inc. · www.flexcompute.com

Research Scientist (2019–2024) → Principal Scientist (2024–2025) → Head of Photonics (2025–2026) → Senior Technical Lead (2026–)

- Lead the effort to deliver physics intelligence through simulation, architecting our multiphysics AI platform and driving the foundational research behind it.
- Senior author of research in which autonomous AI agents, given simulation tools, designed photonic devices with state-of-the-art performance.
- As Head of Photonics, led the department of product teams building our electromagnetic simulation products.
- Architected and built Tidy3D's automatic differentiation engine, applying the same gradient technique behind neural-network training to full 3D Maxwell's solves and enabling large-scale inverse design.
- As one of Flexcompute's first employees, built Tidy3D (351★) from the data model up into an industry product, doing a bit of everything with a focus on the Python client, data model, and product engineering.

Graduate Research Assistant Sep 2014 – Aug 2019

Stanford University, Shanhui Fan Group · web.stanford.edu/group/fan/

- Pioneered methods for using photonic hardware to do analog machine learning: optical backpropagation, wave-based recurrent neural networks, and nonlinear optical activation functions.
- Developed and open-sourced tooling for differentiable electromagnetic simulation and adjoint-based photonic inverse design (ceviche, angler, wavetorch).

Machine Learning Intern Jun – Sep 2018

Rasa Technologies · rasa.com/

- Researched text understanding via named-entity recognition; implemented lookup-table matching as a major open-source feature in the Rasa NLU framework.

Junior Software Engineer Jan – Aug 2014

GudTech Inc.

- Full-stack engineering for commercial inventory management software.

EDUCATION

PhD & MS, Applied Physics Sep 2014 – Aug 2019

Stanford University

Thesis: *Adjoint-Based Optimization and Inverse Design of Photonic Devices* · Advisor: Prof. Shanhui Fan

BS, Physics, With Distinction and Highest Honors Sep 2009 – May 2013

University of Michigan

Thesis: *Wafer Reuse for Low-Cost, Thin-Film III–V Photovoltaic Devices* · Advisor: Prof. Stephen Forrest

SELECTED PUBLICATIONS

Kharel, P., Khavasi, A., Chen, X., **Hughes, T.** (senior author). *Autonomous agentic design for photonics.* **arXiv** (2026).
Hughes, T. et al. *Wave physics as an analog recurrent neural network.* **Science Advances** (2019).
Pai, S., Sun, Z., **Hughes, T.** et al. *Experimentally realized in-situ backpropagation for deep learning in nanophotonic neural networks.* **Science** (2023).
Hughes, T. et al. *Training of photonic neural networks through in-situ backpropagation.* **Optica** (2018).
Hughes, T. et al. *Forward-mode differentiation of Maxwell's equations.* **ACS Photonics** (2019).
Hughes, T. et al. *Adjoint method and inverse design for nonlinear nanophotonic devices.* **ACS Photonics** (2018).
Yamilov, A., Skipetrov, S. E., **Hughes, T.** et al. *Anderson localization of electromagnetic waves in three dimensions.* **Nature Physics** (2023).

3,777 citations · h-index 23 · 9 first-author publications (as of April 2026) · full list.

WRITING

Can AI agents autonomously design components on photonic chips?, Flexcompute Engineering (2026). AI agents equipped with an electromagnetics simulator tackle waveguide bends, crossings, splitters, and demultiplexers through iterative simulation.
Designing a photonic chip component with ~45 lines of Python, Flexcompute Engineering (2026). Compact introduction to photonic inverse design using Tidy3D and the adjoint method.

SELECTED PATENTS

Efficient Analog Backpropagation Training Architecture for Photonic Neural Networks (2023).
Simultaneous Measurements of Gradients in Optical Networks (2022).
Training Wave-Based Physical Systems as Recurrent Neural Networks (2022).
Systems and Methods for Activation Functions for Photonic Neural Networks (2022).
Training of Photonic Neural Networks Through In-Situ Backpropagation (2021).

OPEN SOURCE

tidy3d	GPU-accelerated 3D electromagnetics solver with cloud execution and full autodiff.	351★
ceviche	Differentiable frequency-domain electromagnetic simulation (FDFD), JAX/autograd-compatible.	419★
angler	Adjoint-based inverse design for nonlinear nanophotonic devices.	191★
wavetorch	Wave-based analog RNN simulator; physics ↔ RNN correspondence.	542★
neuroptica	Optical neural network hardware simulation with physical imperfections.	271★

SELECTED INVITED TALKS

Building the future of photonic design with machine learning, Frontiers in Optics, Visionary Speaker (2025).
Hardware-accelerated FDTD for large-scale electrodynamics, UW Madison Computing in Engineering Forum (2022).
Training of photonic neural networks through in-situ backpropagation, CLEO (2019).